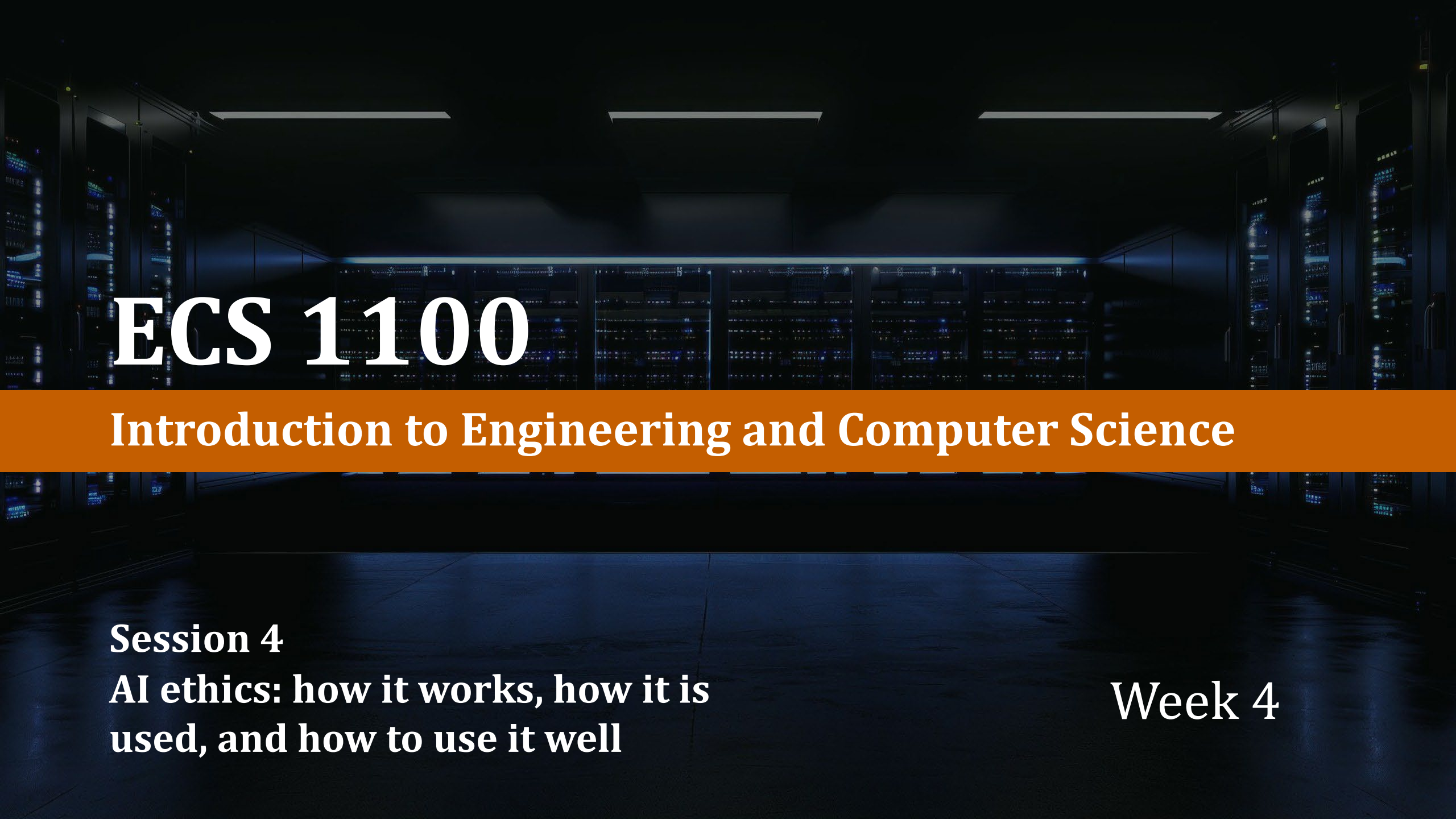


ECS 1100

Introduction to Engineering and Computer Science

Session 4

AI ethics: how it works, how it is used, and how to use it well

Week 4

What comes to mind when you hear AI ethics?

Open Canvas, then Discussions

[ICP A – Week 4]

Post one or two sentences: the first thing that comes to mind, and one example from your own life. 2 points.
There is no wrong answer. You are graded on posting, not on the answer. Posts are visible to the class with your name.

Then we read the board together

Three questions: what repeats, what is missing, which one surprised you. We read this board again at the end of class.

Scan to open the discussion

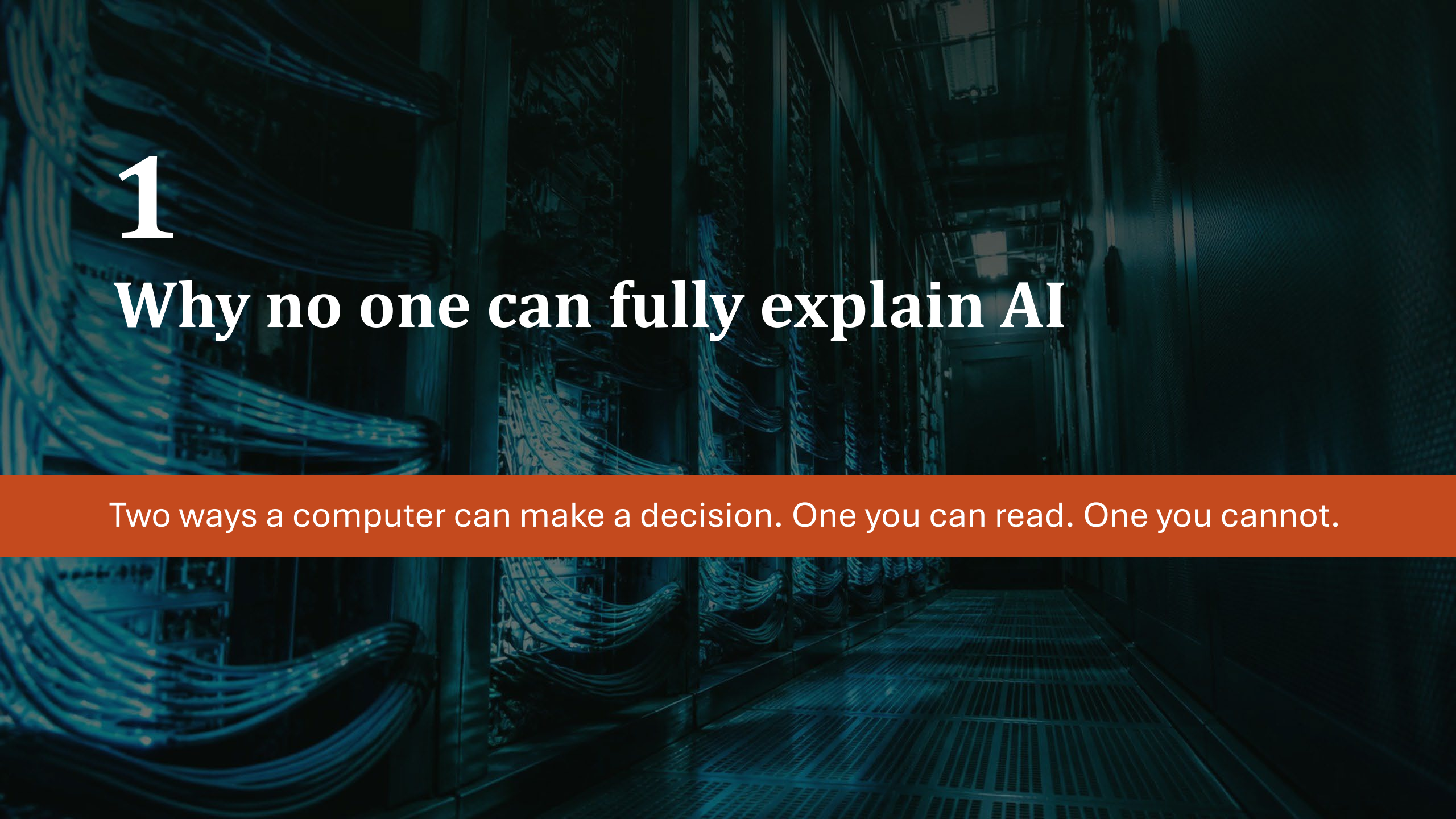
(QR code removed)

What you write is up to you. Say what you actually think, and one example you have met yourself. Short is fine.

THE BOARD

Look at what we posted

What repeats?	Name the worries that appear more than once.
What is missing?	Name a use of AI that nobody posted.
Which one surprised you?	The person who wrote it says one more sentence about it.



1

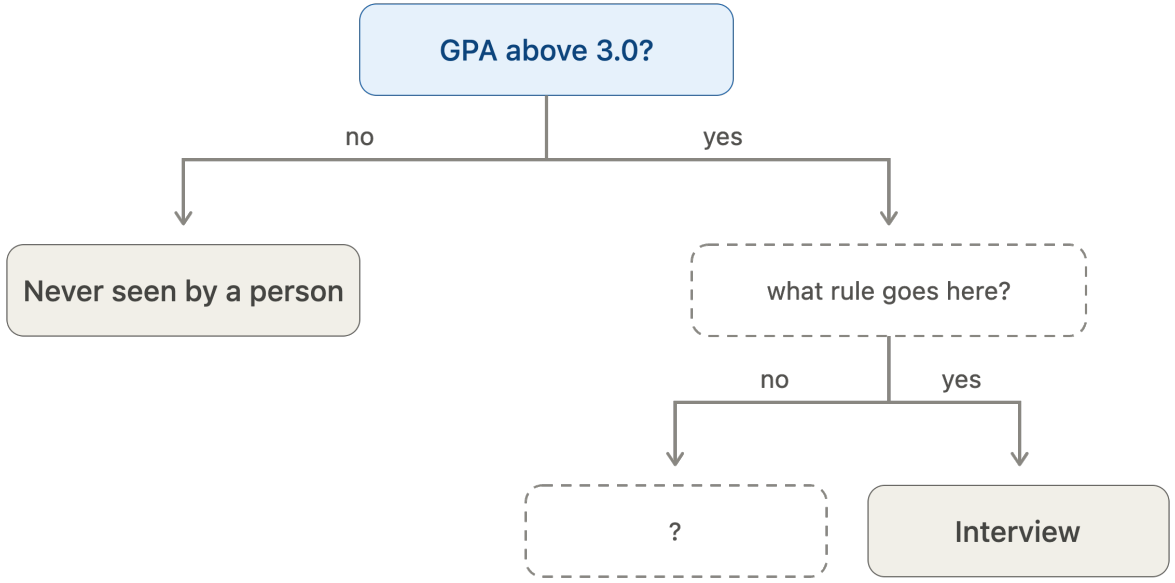
Why no one can fully explain AI

Two ways a computer can make a decision. One you can read. One you cannot.

ONE JOB APPLICATION AND TWO WAYS TO DECIDE IT:

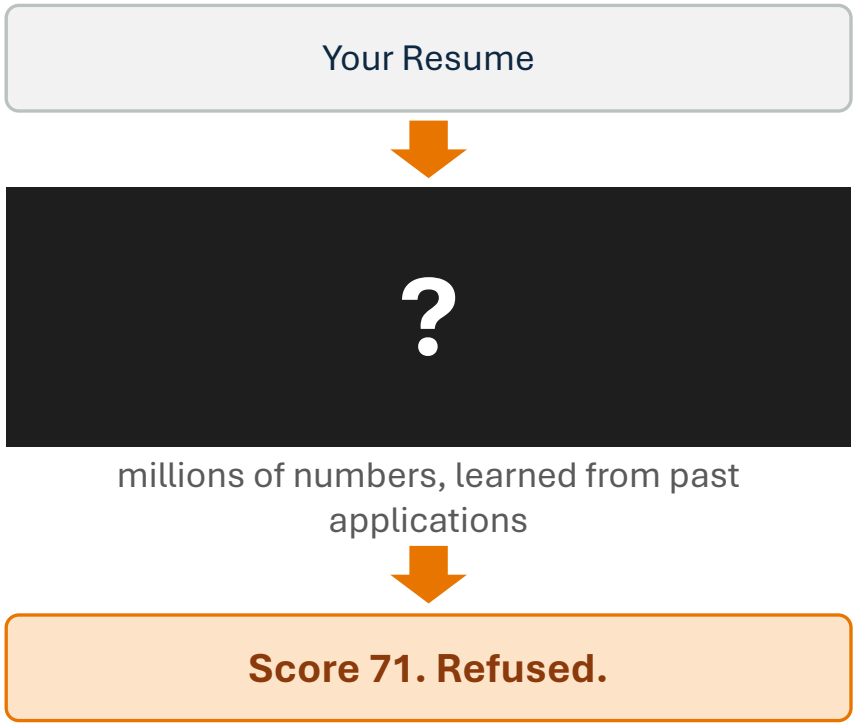
A decision tree, and a black box

A decision tree: rules a person coded



You wrote every box. Someone can ask you why, and you can answer

A black box: patterns nobody wrote



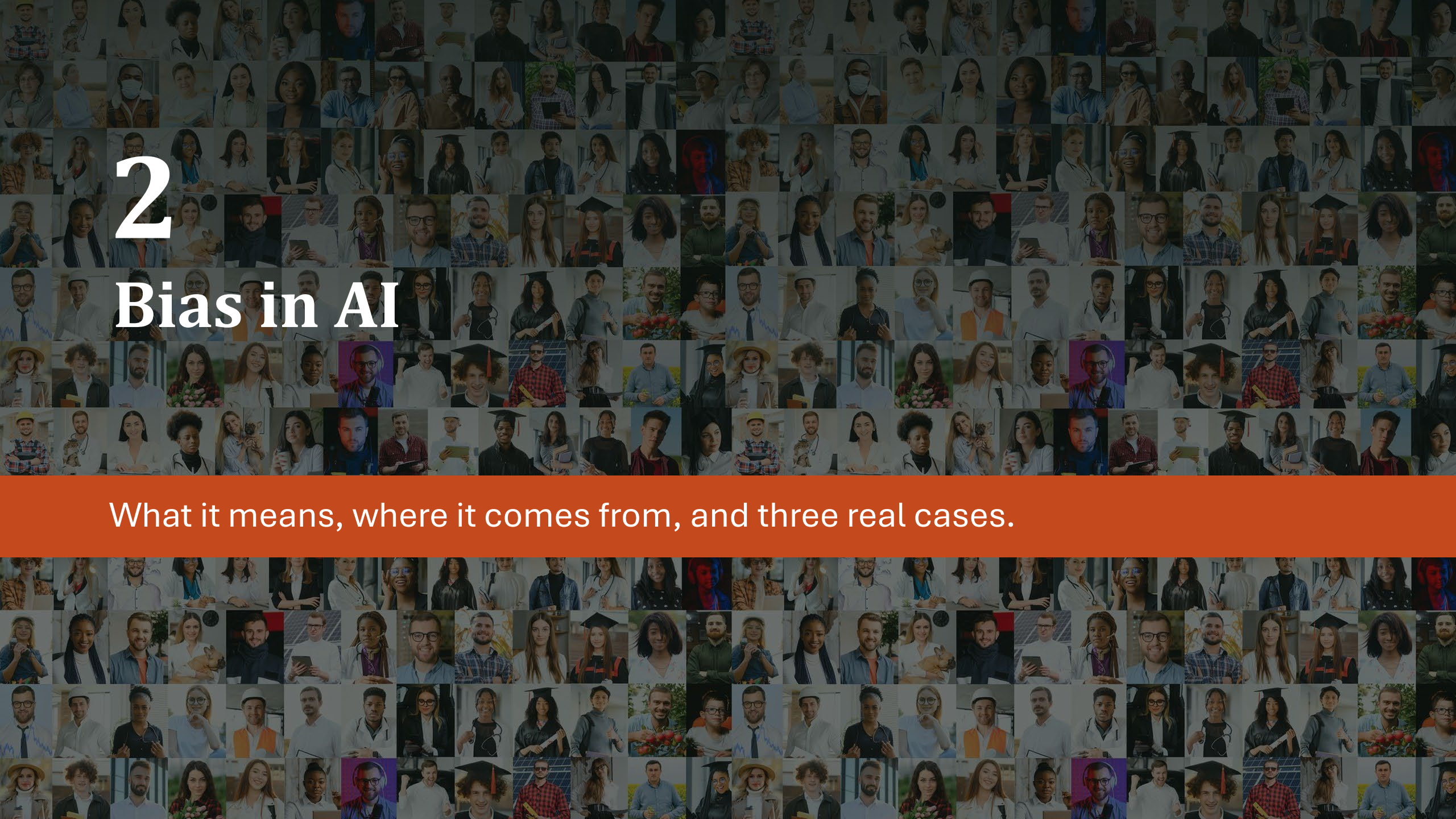
You wrote every box. Someone can ask you why, and you can answer.

No rule to read. Not even the people who built it can say why. A program that decides this way is called a model.

THE AMBIGUITY

Three Blind Spots of the AI test score

Why it answered	There is no reason to read. The same question can get a different answer tomorrow.
Whether it is right	It sounds equally confident when it is wrong. Nothing in the answer tells you which time this is.
Who it works badly for	It was tested on the average. It can be right for most people and wrong for one group, and nobody notices until someone counts.



2
Bias in AI

What it means, where it comes from, and three real cases.

Bias in AI: wrong in one direction

It comes from the examples

The model learns from the past. If the past had few examples of some people, or treated them unfairly, the model learns that too, and repeats it.

It comes from the use

A model built for one place or one group is used on another. The place it is used decides who it is wrong about.

Data bias, in this week's case

The program learned from recordings of students sitting alone in quiet private rooms. That is what it treats as a normal exam.

Deployment bias, in this week's case

It is run on students in shared apartments, with family, roommates and noise. The same movement means something different there, and the program cannot tell.

The model is not prejudiced. It is a mirror of the examples it was given, and of where we point it.

Both kinds are engineering decisions: what to learn from, and where to use it.

THREE REAL CASES · EACH LABELED

Real data, real harm



Hiring · Data bias

Amazon trained a resume-screening model on ten years of resumes it had received. Most came from men. The model learned to mark down resumes containing the word 'women's'. It was scrapped.

Source: Reuters, 10 October 2018



Images · Deployment bias

Image generators trained on the internet produce narrow, stereotyped people. Google built in a correction before launch: show a range. It had no exception for prompts where the range is the wrong answer — historical figures, specific documented moments. The rule was defensible. The missing branch was not.

Source: Google, 23 February 2024



Health · Both bias

The model learned from what care *cost*, not what patients *needed* — that is the data. It was then sold and used as a tool for ranking need — that is the deployment. Some groups had less spent on them for the same illness, and the model read them as healthier.

Source: Science, 25 October 2019



3
How we are using it

The part that matters most in your first semester.

What has an AI already decided for you today?

One minute, with the person next to you. Name three things since you woke up that a program chose, ranked, unlocked, or showed you.

SCHOOL

The AI policy: permitted, and not permitted

Permitted

Brainstorming, checking your understanding, explaining a concept, or feedback on a draft you wrote. Note any tool use at the end of the assignment.

Not permitted

Generating the substance of a reflection, a Connections log, an engagement log entry, or an in-class item.

Fabricating a conversation, an event attendance, or a proof of attendance is a conduct violation, not an AI-policy violation.

This is the AI policy in the syllabus. When you are not sure, ask before you submit. Asking is always safe.

The test in every case: if the AI is not helping you reach the learning outcome, the use is not proper.

CRIMINAL USE

Made-up images and deepfakes

What it is	Software that makes a photo, a voice, or a video of a real person doing something they never did. Minutes, and free.
Why it can be a crime	A fake image of a classmate, a fake voice of a parent asking for money, a fake clip of a public figure. Laws against fraud and harassment apply, and so does the student conduct code.
What to do	Do not make them. Do not share them. Treat any surprising image or clip as unverified until a second source confirms it.
A second source	Not the feed you saw it on: the outlet's own site, the person's own account, a fact-check desk such as AP Fact Check, Reuters Fact Check or Snopes, or a reverse image search. A repost is not a second source.

The person harmed is real even when the picture is not.



4

Four questions, and one story

The engineers' code, four questions, and one story you decide.

The engineers' code of ethics

The National Society of Professional Engineers, NSPE, publishes the code engineers follow. Its six Fundamental Canons, quoted in full from the code as revised in July 2019:

1. Hold paramount the safety, health, and welfare of the public.
2. Perform services only in areas of their competence.
3. Issue public statements only in an objective and truthful manner.
4. Act for each employer or client as faithful agents or trustees.
5. Avoid deceptive acts.
6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession.



The code is what makes engineering a profession and not only a job.

The first one stands above all the others. That is what hold paramount means. The other five do not outrank each other.

Source: NSPE Code of Ethics for Engineers, nspe.org/career-growth/nspe-code-ethics-engineers

Four questions that turn Canon one into a decision

These four are **NOT** a professional standard. The canons are. These are a starting set for turning "hold paramount the safety, health, and welfare of the public" into something you can check in HW assignment. Different situations demand different questions. These four will get you to the right ones.

1

Who could get hurt, and had no say in the decision?

2

What can go wrong, and would anyone notice?

3

What do I have to tell people about it, and who do I have to tell?

4

When the law, my profession's rules and my employer's rules disagree, which one wins?

You will meet these four again in every sector session this semester. They are not about AI only.

ICP B – Week 4

Upload your respective copies to the Canvas.

It should contain your name, role you chose, your arguments (whether keep/change/stop) and WHY

YOU WILL BE OKAY

What happened when calculators arrived



A desk calculator, the machine teachers argued about.

Then Teachers feared the end of arithmetic. What changed was the work: less by hand, more on setting up the problem and checking the answer.

Now The work changes the same way: less producing, more asking the right question, checking the answer, and knowing when not to trust it.

That is what this course is training, and it is the part a black box cannot do for you.

Before you leave

- **Homework 4, five parts, one PDF**
- **One question for the guest speaker**
- **Engagement Category B**

One AI system that has already made a decision about you: find out what it does, answer the four questions about it, take a position, then argue against yourself with a tool. Due Sunday Sep 20, 11:59 pm. About 55 minutes.

Next week is Sector 1, Applied AI and Machine Learning, with a guest speaker. Write the one question you want them to answer. It is part 5 of the template. About 5 minutes.

The Jonsson School ECS Internship and Job Fair, or a University Career Center All-Majors or STEM fair. A resume review appointment or an approved Handshake profile also counts. Log it in the Engagement assignment on Canvas by Friday, September 25, 11:59 pm.

Homework 4 is 30 points, graded for completion and effort, like Homework 2.
Next week: Sector 1, Applied AI and Machine Learning, with a guest speaker. Bring your question.